Unit 1 Quadratic Functions, Equations and Inequalities

Basic

1. The curve $xy + 10 = 0$ and the line $x + 2y + 1 = 0$ intersect at the points P and Q. Find the x-coordinate of P and of Q. [3]

[N21/I/2]

2. Express $3 - 12x - 2x^2$ in the form $a(x + b)^2 + c$ and hence state the coordinates of the turning point of the curve $y = 3 - 12x - 2x^2$. [4]

[N21/I/3]

3. **(a)** Solve the simultaneous equations

$$y = 2x^2 - 7,$$
$$y = 3x + 20.$$ [3]

(b) Find the greatest value of the integer a for which $ax^2 + 5x - 2$ is negative for all x. [3]

(c) Find the values of the constant c for which the line $y = 4x + c$ is a tangent to the curve $y = x^2 + cx + \frac{21}{4}$. [3]

[N20/II/2]

4. Represent the solution set of $15(1 + 2x) \geqslant x(19 - 2x)$ on a number line. [4]

[N20/II/5(a)]

5. Find the set of values of the constant k for which the curve $y = x^2 + (2k + 1)x + 1$ lies entirely above the line $y = x$. [4]

[N19/I/2]

Intermediate

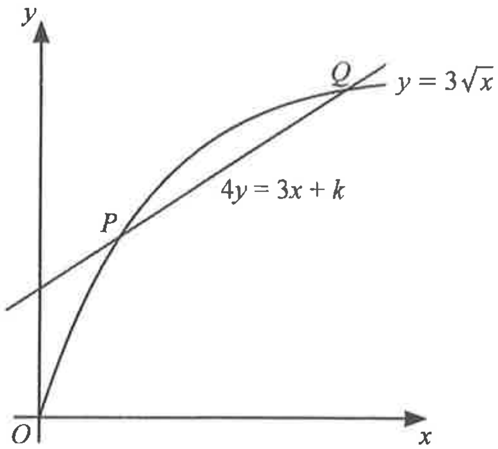
6. A curve has equation $y = 14x - x^2$. Points A and B lie on the curve and have x-coordinates of k and $2k$ respectively, where k is a constant and $0 < k < 7$.

(a) Find the range of values of k for which the y-coordinate of B is greater than the y-coordinate of A. [3]

(b) Explain why the gradient of the curve at A is always greater than the gradient of the curve at B. [2]

[N22/I/4]

7.



The diagram shows the curve $y = 3\sqrt{x}$ and the line $4y = 3x + k$ where k is a constant.

The line intersects the curve at the points P and Q.

(a) In the case where $k = 9$, find the coordinates of P and Q. [4]

(b) In the case where the line is a tangent to the curve, find the value of k. [4]

[N22/I/8]

8. The equation of a curve is $y = 2x^2 - 6x + 3$.

(a) Find the set of values of x for which the curve lies above the line $y = 11$ and represent this set on a number line. [4]

The line $y = 2x + k$ is a tangent to the curve at the point P.

(b) Find the value of the constant k. [3]

(c) Find the coordinates of P. [2]

[N21/II/5]

9. The equation of a curve is $y = 2x^2 + (k + 2)x + k$, where k is a constant.

(i) In the case where $k = 5$, show that the line $y = 19x - 13$ is a tangent to the curve and find the coordinates of the point of contact. [3]

(ii) Explain why there is only one value of k for which y cannot be negative and state this value. [4]

[N18/II/9]

10. The equation of a curve is $y = 9x^2 + (2m + 1)x + 1 + c$, where m and c are constants. The line $y = mx + c$ is a tangent to the curve at the point P.

(i) Find the positive value of m. [4]

(ii) Using this value of m, and given that the curve passes through $(-2, 19)$, find the coordinates of P. [4]

The straight line L meets the curve at one point only.

(iii) Given that L is **not** a tangent to the curve, what can be deduced about L? [1]

[N17/II/6]

11. The equation of a curve is $y = 2x^2 - kx - 4$, where k is a constant, and the equation of a line is $y + 2x = 12$.

(i) In the case where $k = 6$, find the coordinates of the points of intersection of the line with the curve. [3]

(ii) Show that, for all values of k, the line intersects the curve at two distinct points. [2]

[N16/I/1]

12. (i) Given that $ax^2 + 6x + c$ is always negative, what conditions must apply to the constants a and c? [3]

(ii) Give an example of values of a and c which satisfy the conditions found in part **(i)**. [2]

[N15/I/4]